CPU Scheduling Simulator (Round Robin vs SJF)

Project Description

This project is a CPU Scheduling Simulator that compares two scheduling algorithms:

- Round Robin (RR)

- Shortest Job First (SJF)

The system allows users to input multiple processes with arrival time and burst time, then visualizes execution using Gantt charts and calculates performance metrics such as Waiting Time, Turnaround Time, and Response Time.

The main goal is to analyze fairness vs efficiency and study the effect of time quantum in Round Robin scheduling.

Features

- Accept dynamic number of processes

- Input validation (arrival time, burst time, quantum)

- Round Robin scheduling implementation

- Shortest Job First scheduling implementation

- Gantt chart visualization for both algorithms

- Calculation of:

  - Waiting Time (WT)

  - Turnaround Time (TAT)

  - Response Time (RT)

- Comparison between both algorithms

- Error handling for invalid input

How to Run

1. Clone the repository: https://github.com/Anton-M0/CPU-Scheduler-Comparison
git clone

2. Open the project in your IDE (VS Code)

3. Install required dependencies (python)

4. Run the main file: python main.py

Project Structure

src/
model/
scheduler/
metrics/
gui/
util/
screenshots/
test-cases/

Test Scenarios

- Scenario A: Basic workload

- Scenario B: Short job heavy

- Scenario C: Fairness test

- Scenario D: Long job sensitivity

- Scenario E: Validation test

Conclusion

Round Robin is more fair, while SJF is more efficient for short jobs. The quantum value strongly affects Round Robin behavior.

Screenshots

Check the /screenshots folder for:

- GUI

- Gantt Charts

- Results Tables

- Validation Errors